

Pre-Stall Load Redistribution on Swept Tailplane Models with Full-Span Controls

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Local loading changes on a swept lifting surface with a deflected plain control can alter lift growth and pitching moment well before maximum lift. MONNALISA wind-tunnel measurements and Reynolds-averaged Navier–Stokes solutions are used to examine how fixed-surface and full-span control loads combine in this response. For the positive-sweep, zero-dihedral model at a 20° deflection, the computed fixed surface continues gaining lift while both surface contributions fall below their preceding incidence trends. Across the tested baseline windows and zero-incidence solutions, the fixed surface accounts for about 63% of the total lift departure; the control contributes more strongly to the moment departure about the fitted aerodynamic center. Pressure sections and regional integrations connect these different allocations to stronger leading-edge suction, reduced hinge/aft loading, and larger outboard moment weights. Matched dihedral comparisons distinguish absolute load projection from changes with incidence. The detailed interpretation concerns the computed states, for which correspondence with measured integrated coefficients is partial. The combined force and moment analysis identifies loading changes that continuing total lift growth alone would conceal.

Nomenclature

b	=	undeformed model semispan, m
$c, \bar{c}$	=	local chord and mean aerodynamic chord (MAC), m
C_L, C_D	=	lift and drag coefficients
$C_{F,x}, C_{F,z}$	=	body-axis force coefficients
C_M^A	=	pitching-moment coefficient about the respective fitted center
$C_M^O, C_M^{A_I}$	=	moment about numerical origin or common Geom. I fitted axis

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$C_p, \mathbf{C}_f$	=	pressure and vector skin-friction coefficients
$D_{Q,j}$	=	departure of component j from its preceding coefficient trend
f_{rev}	=	specified upper-aft chordwise reversal area fraction
$m_{Q,j}^{\text{pre}}$	=	fitted preceding slope, per degree
$M_\infty, Re_{\bar{c}}$	=	freestream Mach and MAC Reynolds numbers
q_∞, S	=	dynamic pressure and fixed semiplanform reference area
$s, \eta = s/b$	=	undeformed span coordinate and its nondimensional value
u_i	=	supplied propagated uncertainty magnitude at sample i
x_A, x_O	=	numerical fitted-center and original reference abscissae, m
α, δ, Γ	=	displayed incidence, control deflection, and dihedral, deg
λ, Λ, τ	=	quarter-chord sweep, aspect ratio, and taper ratio
ξ	=	local straightened chord fraction
BCM, SA	=	Bas–Cakmakcioglu transition and Spalart–Allmaras closure
RANS, WT	=	Reynolds-averaged Navier–Stokes and wind tunnel

I. Introduction

A deflected trailing-edge control changes both the amount of lift generated by a lifting surface and the location at which that load acts. Its response with incidence is therefore relevant to pitching moment as well as lift, even before the maximum-lift condition is reached. When local loading changes, total lift can continue increasing while its growth slows and the pitching response changes more abruptly. Interpreting this behavior requires relating the integrated coefficients to the loads on the fixed and moving parts of the surface.

Sectional plain-flap experiments establish the starting point for this problem. Sears and Liddell [1] measured lift, drag, pitching moment, and hinge moment for a 30%-chord plain flap on an NACA 0015 airfoil, including sealed and unsealed hinge gaps. Separation reduced flap effectiveness at large deflections, although additional deflection still increased lift within the tested range. Their comparison with a thinner symmetric section also showed that similar lift effectiveness could coexist with different hinge-moment behavior. Reduced flap effectiveness is thus established physics; neither the presence of separation nor a single integrated lift trend fully characterizes the loading response.

On a finite wing, flap-induced loading is also shaped by the spanwise flow and planform. Campbell, Blanks, and Leaver [2] investigated rectangular wings of small aspect ratio with full-span plain flaps of several chord fractions. Their force and pitching-moment measurements, compared with thin-wing theory and empirical thickness and boundary-layer

allowances, related flap effectiveness and center of pressure to flap chord and aspect ratio. Those unswept, thinner, transition-tripped models differ from the present swept models, but they establish why the location of the additional load matters alongside its magnitude.

For swept wings, Harper and Maki [3] documented how spanwise changes in local lift response can produce pitching-moment nonlinearity below maximum lift. The first appearance of local stall need not coincide with the first integrated moment change, because the force change and its moment arm vary along the span. Their broad stall review provides context for examining load redistribution, rather than an equivalent test of the present deflected-control configurations.

These relationships can enter reduced-order models through their aerodynamic inputs and corrections. Sivells and Neely [4] coupled nonlinear sectional lift data to lifting-line theory and obtained improved finite-wing predictions near maximum lift; they also calculated wing pitching moment from sectional moment data. More recently, Goitia and Llamas [5] coupled a vortex-lattice formulation to viscous sectional data, with effective-sweep and finite-wing corrections, and compared nonlinear lift predictions for wing and tail geometries. Such formulations can represent nonlinear responses, although their sectional inputs and three-dimensional corrections govern what they can resolve. The relevant modeling question is therefore whether a particular formulation represents the loading changes of interest, rather than whether its nominal order excludes nonlinearity.

The immediate basis for the present work is the MONNALISA database publication [6]. Its Sec. IV and Fig. 11 already document the baseline pre-stall lift nonlinearity, expansion of reversed flow on the control, associated pressure changes, and continued growth of total lift. The publication also reports nonlinear responses from the DUST lifting-line formulation using sectional tables and corrections, alongside smoother potential-based panel and vortex-lattice predictions. These observations establish both the existing physical interpretation and the model-specific nature of the lower-fidelity response.

The remaining loading question is how changes on the fixed surface and control combine to produce the joint lift and moment response. Continued fixed-surface lift growth, for example, does not establish that this contribution follows its preceding trend. Comparing ordinary increments with departures from that trend identifies which components supply the reduced total growth, while spatial pressure integrations show why the component ranking can differ for moment. This study develops that quantitative interpretation from the existing RANS solutions, first assessing their integrated coefficients against the measured response and its supplied uncertainty. Changes in control deflection and geometry then test how broadly the reference loading pattern persists within the dataset.

The test articles are isolated tailplane-inspired wings with full-span, 30%-chord plain controls; no fuselage, wing wake, or complete-aircraft trim condition is represented. The 45° dihedral setting supplies a pronounced orientation contrast on the same planform, while a second planform changes several geometric parameters together. These comparisons define the scope of the analysis without establishing behavior for partial-span controls, other flap chords,

or intermediate dihedral angles.

II. Experimental, Numerical, and Analysis Framework

A. Models and experimental installation

The study combines measured integrated loads with computed surface loads to connect a pre-stall coefficient change to its spatial contributions. The WT data establish the lift and moment responses; the RANS fields supply the component and regional integrations used to interpret the computed response. Two orientations of the same planform provide the matched dihedral comparison, while a second planform tests the loading pattern under combined geometric changes.

The models have an undeformed semispan $b = 1.6$ m, nominal thickness ratio $t/c = 9\%$, and a full-span plain trailing-edge control rotating about the 70%-chord hinge. Table 1 gives the geometry and global references. Aspect ratio is defined for the corresponding full planform, $\Lambda = 2b^2/S$. Geom. I at $\Gamma = 0^\circ$ and 45° retains the same undeformed planform, area, and MAC. Geom. II simultaneously changes sweep, taper, aspect ratio, area, and chord distribution, so its comparison with Geom. I does not isolate sweep direction.

Table 1 Geometry and reference quantities. Area is the undeformed semiplanform area; $Re_{\bar{c}} = Re_b \bar{c}/b$, with $Re_b = 5.5 \times 10^6$.

Property	I, $\Gamma = 0^\circ$	I, $\Gamma = 45^\circ$	II, $\Gamma = 0^\circ$
λ (deg)	30	30	-15
Λ	5.0	5.0	5.2
τ	0.4	0.4	0.6
S (m ²)	1.024	1.024	0.984604
$\bar{c}$ (m)	0.679184	0.679184	0.6282
Root chord (m, rounded)	0.914286	0.914286	0.76923
$Re_{\bar{c}}$	2.335×10^6	2.335×10^6	2.159×10^6
Numerical x_A (m)	0.642000	0.643532	-0.003618

The WT measurements were obtained in the low-turbulence section of the closed-circuit GVPM facility at Politecnico di Milano. Figure 1 shows the floor-mounted semispan installation. The dummy floor conceals the support, balance, and control actuation mechanism. The documented test-section width and length are 4 and 6 m, respectively; the undeformed semispan is 40% of that width. A consistent installed-height value and a floor-layer profile at the model are unavailable for a quantitative confinement assessment. Thus neither a thin floor layer nor negligible wall effects is assumed.

The database setup description [6] gives freestream turbulence below 0.1%, measured using a turbulence sphere and hot-wire anemometry. Outside the wall layers, the inlet total-pressure variation was within $\pm 0.5\%$ of dynamic pressure over most of the section and within $\pm 1\%$ overall; outlet measurements confirmed similar uniformity. The reported longitudinal static-pressure gradient is $\partial C_P / \partial x = -0.004 \text{ m}^{-1}$. Loads were acquired through a RUAG 767 strain-gauge balance below the floor and an HBM QuantumX system; the reduction includes the balance-to-wind transformation and

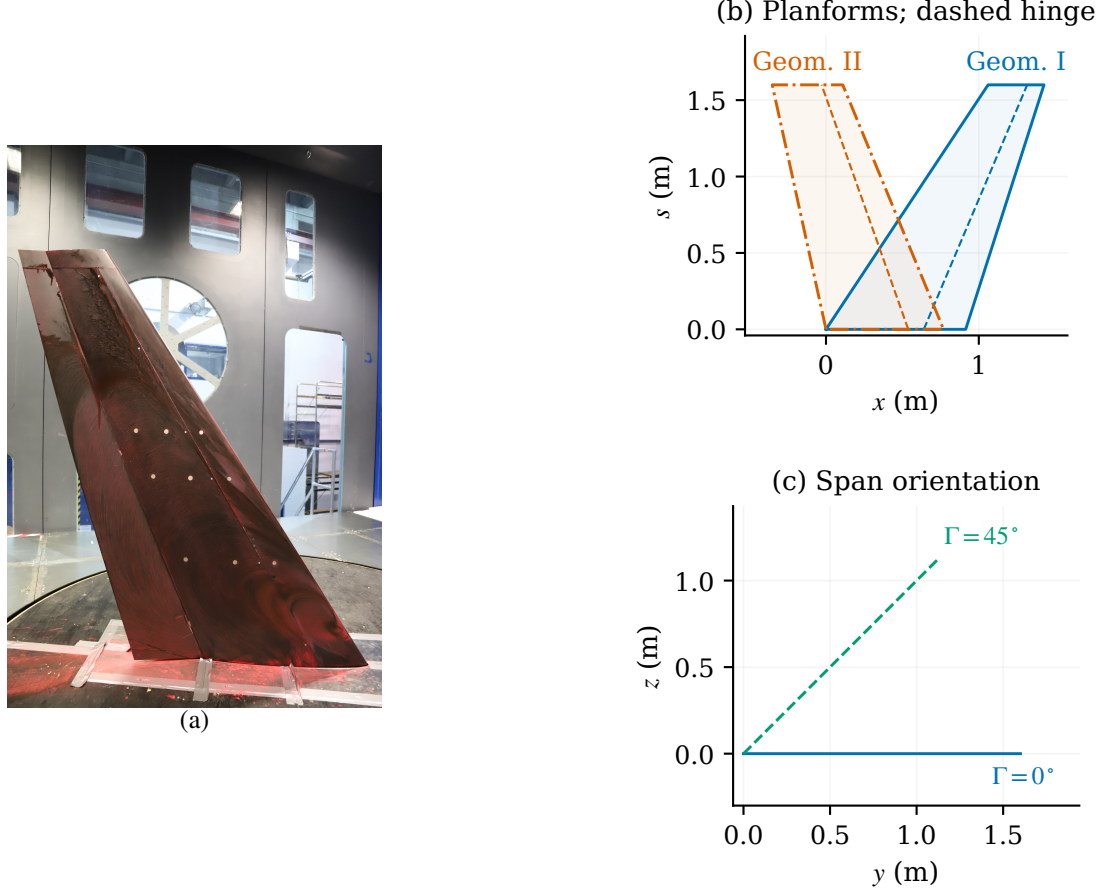


Fig. 1 Model installation and geometry: (a) oil-flow photograph at $\alpha = 25^\circ$, $\delta = 20^\circ$, illustrating the GVPM mounting rather than the pre-stall flow state; (b) undeformed planforms and 70%-chord hinges; (c) metric span orientation in numerical axes, with $b = 1.6$ m at both dihedral settings. Dimensions are given in Table 1.

the database corrections. Nominal conditions are $M_\infty = 0.147$, $T_\infty = 300$ K, and the MAC Reynolds numbers in Table 1. The span-based database Reynolds number is retained only for traceability.

The measurements use the LE1 profile variant. Recorded incidences are $-25, -24, -22, \dots, 22, 24, 25^\circ$: the local pre-stall spacing is 2° , with 1° endpoint steps. The comparisons use $\delta = 0, 10, 20, 30^\circ$. Uncertainty bars are the supplied propagated measurement/reduction-chain magnitudes, including balance and transformation contributions. Each fitted-center moment retains its associated propagated uncertainty. A common coverage factor and between-incidence covariance are unavailable; the experimental contrast sensitivities below state their assumptions explicitly.

B. Conventions and moment references

In the numerical body axes, x is downstream, y is spanwise, and z is upward. The force rotation and normalization are

$$C_L = C_{F,z} \cos \alpha - C_{F,x} \sin \alpha, \quad C_D = C_{F,z} \sin \alpha + C_{F,x} \cos \alpha, \quad C_{F,k} = F_k / (q_\infty S). \quad (1)$$

Pitching moment is positive nose-up and normalized by $q_\infty S \bar{c}$. Table 2 defines the angle and frame mapping used to express each measured or numerical state in the displayed convention.

Table 2 Conversion to the displayed convention. Subscript r denotes the existing reduced WT record; n denotes a native numerical state.

Source	Displayed quantities and state handling
Reduced WT	$\alpha = -\alpha_r$, $\delta = \delta_r$, $C_L = -C_{L,r}$, $C_M^A = -C_{M,r}^A$; drag and uncertainty magnitudes retain their signs.
RANS, direct	Native α_n , δ_n retained; forces use Eq. 1.
RANS, $\Gamma = 0^\circ$ reflected	Opposite-deflection parent gives $\alpha = -\alpha_n$, $\delta = -\delta_n$. Positions/forces transform with $T = \text{diag}(1, 1, -1)$; moments with $(\det T)T$; physical sides exchange.
RANS, $\Gamma = 45^\circ$	Signed incidence is directly computed; no same-dihedral reflection.
Zero incidence	Direct and symmetry-reflected zero-incidence solutions remain distinct. Each field uses the force state of its corresponding solution.

The principal WT–RANS comparison uses C_M^A , the moment about each dataset’s zero-deflection fitted aerodynamic center. Following the database construction [6], linear fits to zero-deflection moment and force over $|\alpha| \leq 10^\circ$ define a fixed reference for all control deflections. The WT reduction uses uncertainty-weighted fits, and the numerical references use the retained historical linear fits. This construction removes the fitted zero-deflection moment variation associated with the reference location and provides a consistent basis for examining the deflected-control response. The WT and RANS axes are fitted separately, however, and need not coincide physically; neither is an instantaneous aerodynamic center or an imposed quarter-MAC location. Without physical registration between the models, their moments remain at these respective axes.

The original numerical reference is $O = (0.228571, 0, 0)$ m, the root quarter-chord of Geom. I. For a numerical pole displaced only in x ,

$$C_M(x_1) = C_M(x_0) + \frac{x_1 - x_0}{\bar{c}} C_{F,z}. \quad (2)$$

The fitted abscissae are in Table 1. Matched dihedral diagnostics use the identical numerical axis $A_I = (0.642000, 0, 0)$ m in both configurations, denoted $C_M^{A_I}$. Geom. II uses its own MAC once for every moment contribution; all coefficients use the fixed areas in Table 1.

C. Numerical method and existing accuracy evidence

The existing SU2 [7] calculations solve compressible ideal-gas RANS at the nominal WT Mach and Reynolds conditions, with Sutherland viscosity, SA turbulence closure [8], and BCM transition treatment [9]. The BCM closure supplies an algebraic intermittency correction rather than an independently transported transition equation. Unstructured finite volumes use median-dual vertex control volumes. Mean-flow convection uses Roe fluxes with second-order MUSCL reconstruction [10], Green–Gauss gradients, and a Venkatakrishnan limiter [11] with coefficient 0.05. SA

convection uses scalar upwinding without MUSCL, hence first-order reconstruction. The stated spatial orders therefore apply separately to the mean-flow and SA equations.

The surfaces have no-slip, zero-heat-flux boundaries; the exterior uses a far-field boundary and the root plane a symmetry boundary. This excludes the WT floor, its layer, and the support. Implicit Euler pseudo-time advancement, FGMRES linear solves, adaptive CFL, and staged restarts were used to obtain steady states. The configured Cauchy stopping criterion monitors lift and drag over 100 entries, commonly at 3×10^{-6} and at 10^{-6} in later Geom. II stages. Moment is not a stopping quantity; iteration caps and restarts also occur.

The saved surface states are checked against their force reports and terminal histories. Matches are established for 41 of the 44 interpreted states; the exceptions and incomplete final-stage reconstruction-order registration are documented in Appendix A. These checks establish the available state correspondence, while the grid assessment and measured coefficients address distinct aspects of numerical accuracy.

The automated mesh procedure [6] uses anisotropic surface triangles, leading-edge and hinge/junction refinement, a hex-dominant octree volume, and incidence-oriented wake refinement. Nominal controls are a 2.5° leading-edge angle, 1° junction angle, seven trailing-edge points, maximum surface spacing 0.02 m, and boundary-layer growth ratio 1.15. Actual first-cell height and layer count are not registered for every retained mesh. Representative stored y^+ medians are 0.26 for $I/\delta = 20^\circ/\alpha = 2^\circ$, 0.19 for $I/\Gamma = 45^\circ/\delta = 30^\circ/\alpha = 0^\circ$, and 0.23 for $II/\delta = 20^\circ/\alpha = 0^\circ$, with maxima approximately 1.22, 1.26, and 1.13. These support resolved near-wall treatment for those samples.

Selected surface meshes have 122,860 points/182,423 elements for $I/\delta = 20^\circ$ at 0, 2, 4° , 141,828/209,641 for $I/\Gamma = 45^\circ/\delta = 30^\circ$ at -2, 0, 2° , and 105,506/156,333 for $II/\delta = 20^\circ$ at 0, 2, 4° . These are surface counts. The retained volume for the $I/\alpha = 0^\circ/\delta = -20^\circ$ parent contains 9,440,030 points and 18,918,212 volume elements; equivalent registered volume counts are not available for every interpreted state.

The nominal mesh level was selected for the extensive MONNALISA campaign, balancing spatial resolution with the computational demands of covering multiple geometries, incidences, and control-surface deflections. The choice was informed by the four-level lift/drag refinement assessment in Ref. [6], summarized in Table 3 and Fig. 2. For Geom. I at $\delta = 35^\circ$, $\Gamma = 0, 45^\circ$, and $\alpha = 0, 18^\circ$, the nominal grids contain approximately 9.7–12.7 million points, compared with 50.1–65.9 million on the finest level. Nominal-to-finest differences are 1.20–2.14% for lift and 1.13–2.49% for drag. This evidence informed the nominal-level production choice while maintaining broad parameter-space coverage. Pitching moments subsequently recovered from the same existing calculations add a condition-dependent sensitivity assessment: the C_M^O difference at $\Gamma = 0^\circ$, $\alpha = 18^\circ$ is 0.01343, and the inclined sequence is nonmonotonic.

The assessment characterizes integrated-load sensitivity at these database conditions; it does not quantify spatial-discretization uncertainty in the local pre-stall flow or component contributions. The experimental comparison in Sec. III provides a complementary assessment: the nominal-grid reference calculation reproduces a lift-slope reduction and a change toward less-negative pitching moment, with a displaced lift interval and differences in coefficient level.

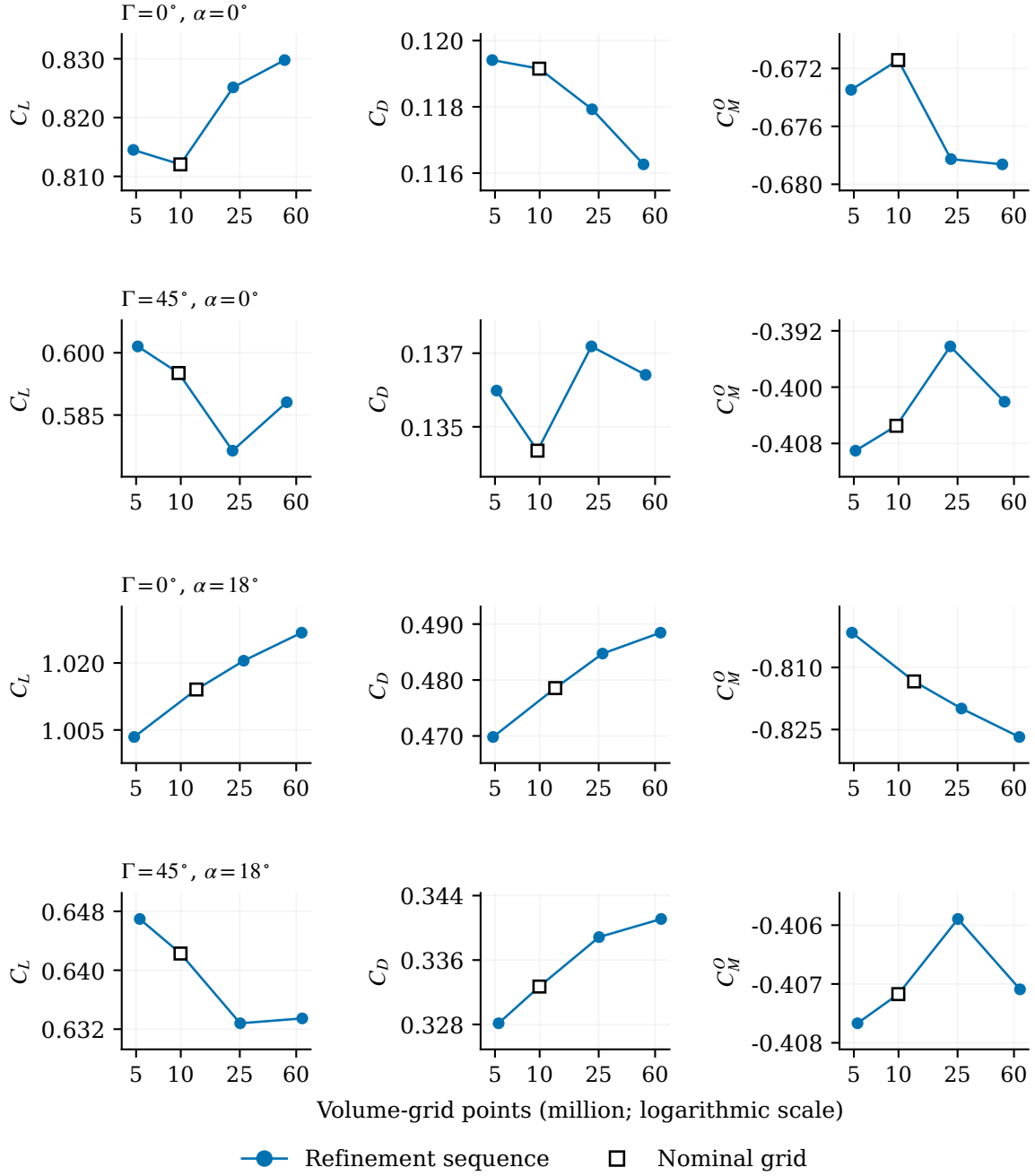


Fig. 2 Existing four-level Geom. I study at $\delta = 35^\circ$: rows show $(\Gamma, \alpha) = (0, 0), (45, 0), (0, 18), (45, 18)^\circ$; columns show C_L , C_D , and C_M^O . The abscissa is the actual volume-grid point count on a logarithmic scale; left-to-right samples are coarse, nominal, fine, and very fine. Open squares identify nominal grids. Moments use $O = (0.229, 0, 0)$ m and the Geom. I area/MAC. The two finest $\alpha = 0^\circ$ moments are taken from the same terminal histories as lift and drag; the other moments use matching force reports. Volume-element counts range from 9.51 to 109.98 million. Lines connect retained refinement levels, without an asymptotic-convergence estimate.

Table 3 Grid-size ranges of the existing four-level study. Counts are volume-grid points and volume elements, in millions. Each range spans Geom. I at $\Gamma = 0, 45^\circ$, $\alpha = 0, 18^\circ$, and $\delta = 35^\circ$; Geom. II is not part of this study.

Refinement	Points	Volume elements
Coarse	4.79–5.29	9.51–10.90
Nominal	9.67–12.74	19.06–24.07
Fine	22.33–26.56	39.93–45.70
Very fine	50.11–65.89	87.35–109.98

D. Additive loading and empirical departures

A total lift increase can conceal reduced growth of one or more surface contributions. The analysis therefore compares the actual coefficient change with the change obtained by continuing an empirical preceding trend, then resolves both measures into additive loads. Solver surface groups define the fixed surface (upper/lower fixed panels), the control (upper/lower control panels, hinge junction, and blunt trailing edge), and tip. Pressure and viscous contributions are retained on both physical sides with the same global S and $\bar{c}$. Tip is included in every total, and all component coefficients are displayed at their actual magnitude.

For coefficient Q and component j , the ordinary increment measures the loading change between the two sampled incidences. Subtracting the continuation of its preceding slope gives the departure,

$$\Delta C_{Q,j} = C_{Q,j}(\alpha_b) - C_{Q,j}(\alpha_a), \quad D_{Q,j} = \Delta C_{Q,j} - m_{Q,j}^{\text{pre}}(\alpha_b - \alpha_a). \quad (3)$$

The slope is an unweighted linear least-squares fit to the declared preceding samples. Using identical samples and weights for every component and the total preserves additivity for both increments and departures. The primary $\delta = 20^\circ$ baseline is $\{-6, -4, -2\}^\circ$; the early $I/\delta = 30^\circ$ numerical baseline is $\{-16, -14, -12\}^\circ$. These are empirical references, with no assumption that the preceding flow is attached. Adjacent windows and the direct and symmetry-reflected zero-incidence solutions test the dependence of the result on that reference. Where the preceding sequence is curved, signed finite differences describe the response without imposing a unique event amplitude.

Measured departures must also be assessed against the supplied uncertainty. For WT contrasts, Eq. 3 is written as $D = \sum_i w_i C_i$, combining repeated uses of any sample into one weight. The two reported sensitivities are

$$u_{\text{ind}} = \left[\sum_i (w_i u_i)^2 \right]^{1/2}, \quad u_{\text{L1}} = \sum_i |w_i| u_i. \quad (4)$$

The first is conditional on independence; the second bounds a perturbation when each sample varies within its supplied magnitude. Neither is an established confidence interval or a new experimental uncertainty budget. Covariance-dependent significance is not inferred.

The component budgets identify which surface contributes, but pressure integration is needed to locate the changes within it. Saved nodal C_p and vector $\mathbf{C}_f$ are therefore integrated over the complete triangulated surface. Linear traction and position-times-traction are integrated exactly within each triangle, with

$$C_M(x_A) = \frac{1}{q_\infty S \bar{c}} \int [z dF_x - (x - x_A) dF_z]. \quad (5)$$

The two terms are retained separately where a common-axis dihedral comparison needs them. All 44 full-surface reconstructions have endpoint residuals below 1.5×10^{-4} . Maximum lift and moment residuals are 2.71×10^{-5} and 7.06×10^{-5} . Conservative absolute-weight residual bounds are checked separately against each increment or departure, with a 10% effect-scale limit for an adopted interpretation. For the baseline lift departure, the bound is 1.01×10^{-5} against $|D_L| = 0.11060$. Small shear changes that fail their corresponding bound are not assigned a resolved mechanism. Accounting closure and reconstruction agreement are distinct from iteration, mesh sensitivity, and experiment–model discrepancy; global residuals do not bound each regional error.

Geometric bins use $\eta = s/b$ and straightened ξ , obtained by undoing the control rotation about the three-dimensional hinge. They include both sides and root extensions; end faces are retained separately. Whole triangles are assigned by centroid. These geometric regions differ from the solver surface groups, especially in their assignment of the hinge junction. Sectional pressure curves use $\eta = 0.2, 0.5, 0.8$ and preserve actual contour connectivity on the native upper/lower fixed and control panels, excluding junction and end faces. Three sections illustrate pressure changes; complete integrations supply the budgets.

A chordwise reversal indicator measures the actual upper geometric-aft area ($\xi \geq 0.7$, excluding end caps) on which $\mathbf{C}_f \cdot \mathbf{t}_c < -\epsilon$, with downstream chord direction projected into the face tangent. Linear nodal signs are clipped within triangles. The fraction f_{rev} is tested at $\epsilon = 0, 10^{-6}, 10^{-5}$. It measures a specified component of reversed shear, rather than the full three-dimensional separation footprint.

III. Measured Response and Numerical Correspondence

The experimental comparison first establishes whether lift and moment change over the same incidence interval and how that response varies with deflection. Figure 3 presents the measured Geom. I baseline polars, including the lower 10° deflection. Figure 4 extends the comparison to both dihedral settings and the second planform at their respective fitted moment references. Its separate strips locate selected WT intervals; Table 4 assesses the corresponding departures using the supplied uncertainty.

For $I/T = 0^\circ/\delta = 20^\circ$, the measured lift slope decreases sharply over $-2 \rightarrow 0^\circ$: lift grows by 0.0199, compared with 0.0892 over $-4 \rightarrow -2^\circ$. Over the same interval, moment increases by 0.0360, becoming less negative while retaining its sign. Both departures exceed the stated uncertainty sensitivities (Table 4); the corresponding pointwise lift- and

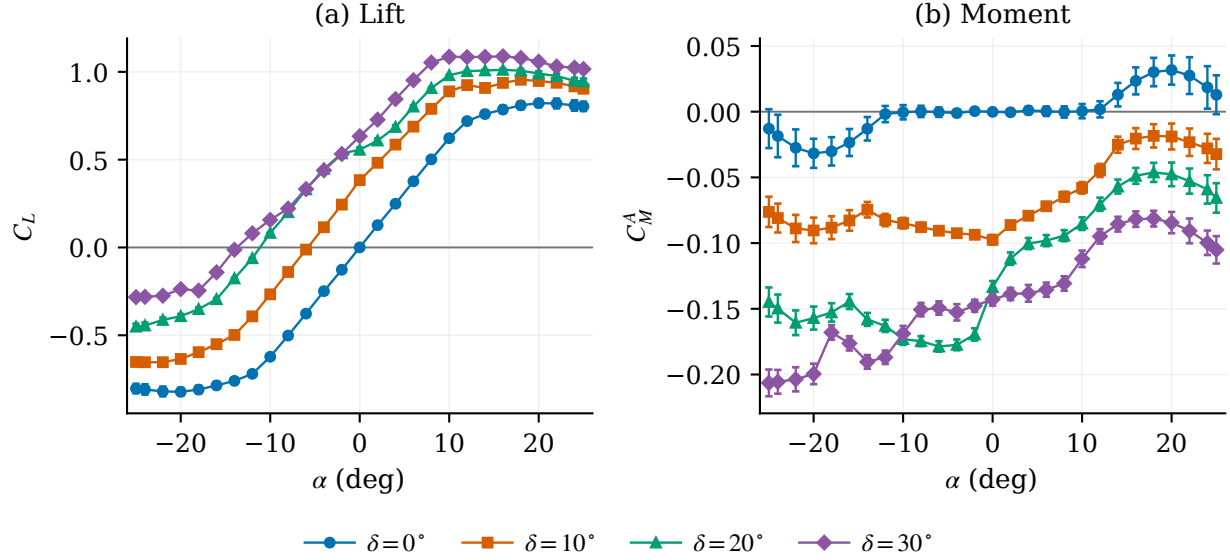


Fig. 3 LE1 WT response of Geom. I, $\Gamma = 0^\circ$, for $\delta = 0, 10, 20, 30^\circ$. Moment C_M^A uses the experimental zero-deflection fitted center, held fixed across deflection. Bars are the paired supplied propagated magnitudes, without an assigned common coverage factor.

Table 4 Primary LE1 WT departure contrasts. Each baseline is $\{-6, -4, -2\}^\circ$; brackets are selected separately for lift and moment. Moment uses the WT fitted center. u_{ind} and u_{L1} are the full-contrast sensitivities of Eq. 4, not confidence limits.

Geom., Γ, δ	Coefficient	Interval (deg)	D	u_{ind}	u_{L1}
I, $0^\circ, 20^\circ$	C_L	$-2 \rightarrow 0$	-0.08274	0.02357	0.03631
I, $0^\circ, 20^\circ$	C_M^A	$-2 \rightarrow 0$	+0.03148	0.00862	0.01360
I, $45^\circ, 20^\circ$	C_L	$-2 \rightarrow 0$	-0.04785	0.01694	0.02634
I, $45^\circ, 20^\circ$	C_M^A	$-2 \rightarrow 0$	+0.00847	0.02630	0.04082
I, $45^\circ, 30^\circ$	C_L	$-2 \rightarrow 0$	-0.04038	0.01646	0.02996
I, $45^\circ, 30^\circ$	C_M^A	$-4 \rightarrow -2$	-0.02860	0.02478	0.03822
II, $0^\circ, 20^\circ$	C_L	$0 \rightarrow 2$	-0.06902	0.02090	0.03739
II, $0^\circ, 20^\circ$	C_M^A	$0 \rightarrow 2$	+0.01668	0.00793	0.01424

moment-coefficient uncertainty magnitudes are approximately 0.0127–0.0133 and 0.0046–0.0047. These sampled contrasts identify the response change at the available incidence spacing.

The lower-deflection result shows why the two coefficients should be examined together. At 10° , the measured moment increment changes from -0.0038 over $-2 \rightarrow 0^\circ$ to $+0.0113$ over $0 \rightarrow 2^\circ$, so a moment-response change is present below the deflection producing the pronounced 20° lift slowdown.

Increasing deflection to 30° instead gives a broader early feature: lift increments are 0.0765, 0.0642, and 0.1113 over $-12 \rightarrow -10$, $-10 \rightarrow -8$, and $-8 \rightarrow -6^\circ$, while moment already increases by about 0.0180 and 0.0183 on the first two intervals. Curvature in the preceding lift sequence makes its departure window-sensitive, spanning both signs.

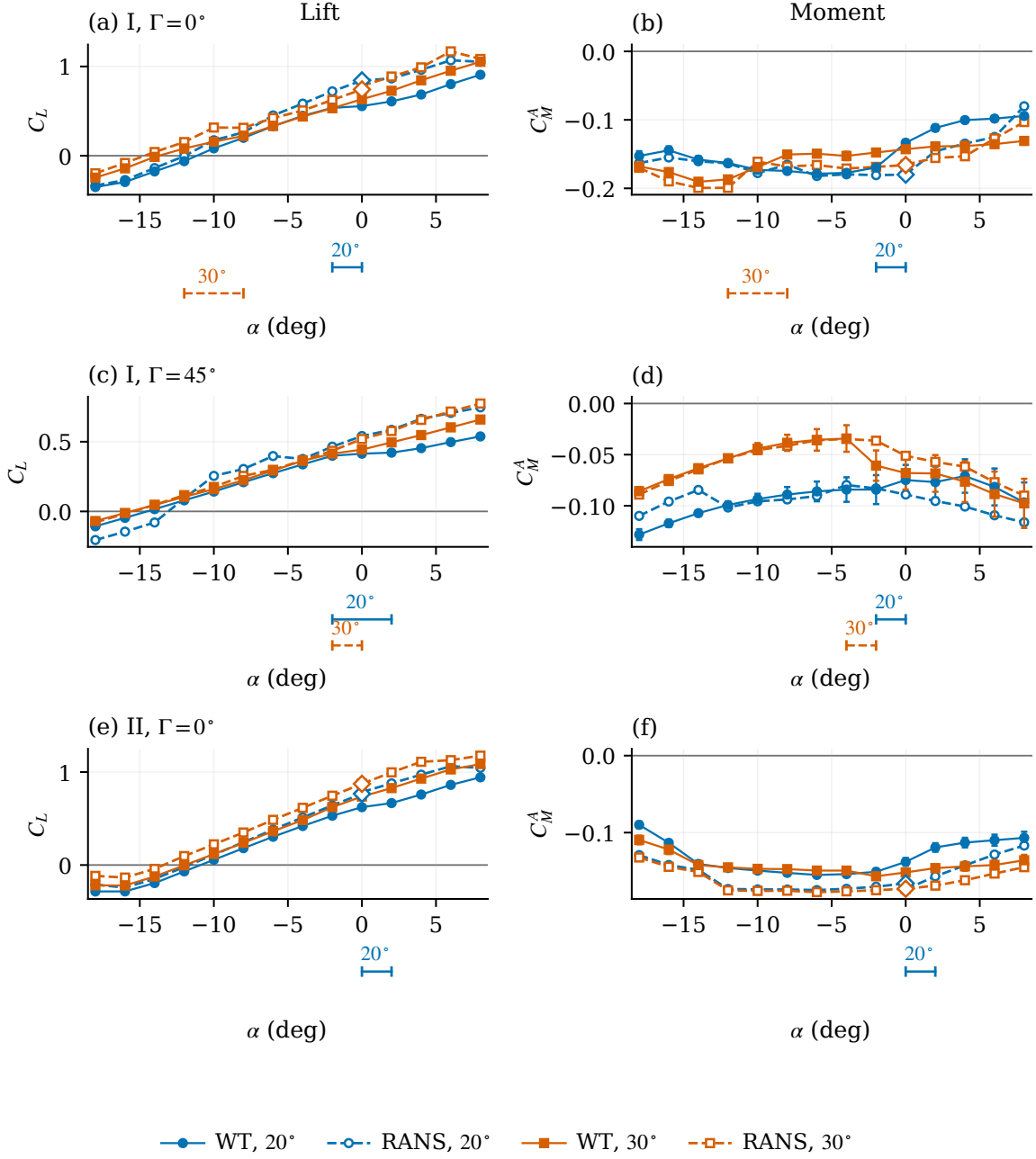


Fig. 4 WT-RANS comparison of lift and moment for $I/\Gamma = 0^\circ$ (a,b), $I/\Gamma = 45^\circ$ (c,d), and $II/\Gamma = 0^\circ$ (e,f). Filled/solid: LE1 WT; open/dashed: RANS. Circles and squares distinguish $\delta = 20^\circ$ and 30° . Separate strips show selected WT intervals, subject to the uncertainty discussion. Open diamonds identify symmetry-reflected zero-incidence solutions. Moment uses each dataset's zero-deflection fitted center; direct and reflected branches remain separate.

Local increments therefore describe this feature more directly than one departure amplitude.

Changing orientation on the same planform does not simply shift this joint response. For $I/\Gamma = 45^\circ/\delta = 20^\circ$, measured lift growth remains suppressed across $-2 \rightarrow 2^\circ$, but the positive moment increment over $-2 \rightarrow 0^\circ$ has a departure smaller than both uncertainty sensitivities. At $\delta = 30^\circ$, the selected lift slowdown occurs over $-2 \rightarrow 0^\circ$, whereas moment decreases earlier, by 0.0265 over $-4 \rightarrow -2^\circ$. That moment departure is comparable with u_{ind} and smaller than u_{L1} . Thus coincident, clearly resolved lift and moment events cannot be assumed for the inclined model.

The second planform provides a further contrast in the joint response. For $II/\delta = 20^\circ$, measured lift grows by 0.0440 over $0 \rightarrow 2^\circ$, compared with approximately 0.0921 on the preceding and following intervals, and moment increases by 0.0187. The primary moment departure exceeds both sensitivities, although adjacent-window departures do not uniformly exceed the primary L1 magnitude. Its lift and moment changes are locally evident, with the 30° response also shown in Fig. 4 for context.

To assess what the surface fields can explain, the numerical response is compared separately for lift and moment. For $I/\delta = 20^\circ$, the RANS lift slowdown occurs over $0 \rightarrow 2^\circ$, one sampled interval later than in WT. At zero incidence the direct and reflected numerical lift coefficients are approximately 0.8422 and 0.8434, compared with WT 0.5560 and uncertainty magnitude 0.0133. A qualitatively similar slowdown and less-negative moment response therefore coexist with substantial lift-level bias. At 30° , the early numerical lift change is sharper than the broad measured feature, and its fitted-center moment departure is window-sensitive.

At $\Gamma = 45^\circ$, numerical $\delta = 20^\circ$ lift decreases over $-6 \rightarrow -4^\circ$ while WT increases; the computed sequence does not reproduce the sustained measured slowdown or weak moment response convincingly. At $\delta = 30^\circ$, a negative moment change is present, with the strongest numerical bracket later than the measured $-4 \rightarrow -2^\circ$ interval. For $II/\delta = 20^\circ$, the direct numerical $0 \rightarrow 2^\circ$ lift increment is 0.0935, more than twice WT 0.0440. The corresponding fitted-center moment increment, 0.0168, is locally close to WT 0.0187, retaining the distinct reference constructions.

The reference calculation captures the direction of the joint response, but the other comparisons show that this correspondence is not uniform. The following analysis therefore asks how the computed component loads produce their own integrated lift and moment changes, using pressure and reversal as supporting numerical evidence. It does not validate a measured local topology. The symmetry-root/floor distinction remains a modeling limitation, without the integrated comparisons assigning the discrepancies to tunnel effects.

IV. Joint Numerical Loading Response

A. Ordinary growth and reduced growth of the fixed surface

For the reference RANS sequence, the continuing total lift increase is supplied by a gain on the fixed surface that outweighs a loss on the control (Fig. 5 and Table 5). This occurs for $I/\Gamma = 0^\circ/\delta = 20^\circ$ over direct $0 \rightarrow 2^\circ$. The positive

fixed-surface increment, however, is smaller than continuation of its preceding slope would give. Relative to the shared baseline $\{-6, -4, -2\}^\circ$, both components have negative departures and the total is $D_L = -0.1106$. The reduction in total lift growth consequently involves the fixed surface as well as the control, despite their opposite ordinary increments.

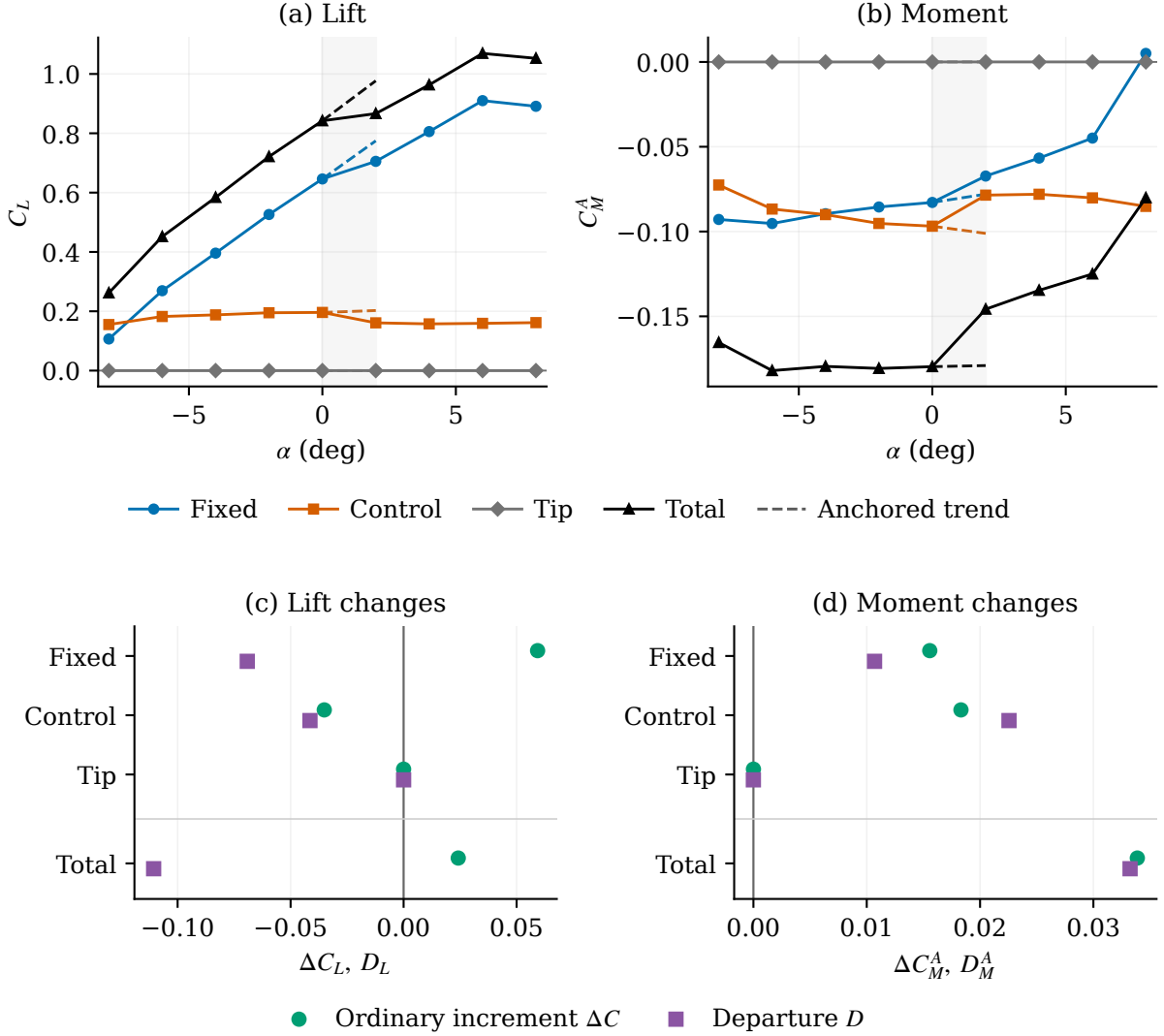


Fig. 5 $I/\Gamma = 0^\circ/\delta = 20^\circ$: component coefficients (a,b) and signed ordinary-increment/departure pairs (c,d). Dashed segments continue the fitted preceding slopes from the direct zero-incidence values; their endpoint gaps equal the departures in Eq. 3. Slopes use $\{-6, -4, -2\}^\circ$; shading marks direct $0 \rightarrow 2^\circ$. All components, including tip, use global area/MAC and fitted axis $x_A = 0.642$ m. Total is separated from the component rows; markers represent signed contributions, not uncertainty intervals.

Moment about the fitted aerodynamic center gives a different component ranking. Both moment departures are positive, and the control contributes more strongly (Table 5). Thus the fixed surface supplies the larger lift-growth deficit while the control supplies the larger moment departure, even though both are formed from the same loading states. The difference motivates examination of where the pressure changes occur and how they are weighted in moment.

Table 5 Numerical ordinary increments and departures, including tip, for zero-dihedral models. Moments use each geometry’s fitted aerodynamic center (Table 1). Direct zero-incidence solutions are used. Primary baselines: $\{-6, -4, -2\}^\circ$ for $\delta = 20^\circ$, $\{-16, -14, -12\}^\circ$ for the early I/ $\delta = 30^\circ$ interval. Six-decimal rounding can affect the displayed sum.

Geom., δ , interval	Measure	Fixed	Control	Tip	Total
I, 20° , $0 \rightarrow 2^\circ$	ΔC_L	+0.059287	-0.035104	-0.000004	+0.024179
I, 20° , $0 \rightarrow 2^\circ$	D_L	-0.069179	-0.041428	+0.000004	-0.110605
I, 20° , $0 \rightarrow 2^\circ$	ΔC_M^A	+0.015562	+0.018314	+0.000002	+0.033878
I, 20° , $0 \rightarrow 2^\circ$	D_M^A	+0.010689	+0.022553	-0.000004	+0.033238
I, 30° , $-10 \rightarrow -8^\circ$	ΔC_L	+0.040586	-0.043560	-0.000002	-0.002974
I, 30° , $-10 \rightarrow -8^\circ$	D_L	-0.067327	-0.052888	-0.000002	-0.120216
I, 30° , $-10 \rightarrow -8^\circ$	ΔC_M^A	-0.014959	+0.007907	+0.000002	-0.007050
I, 30° , $-10 \rightarrow -8^\circ$	D_M^A	-0.017339	+0.015008	+0.000002	-0.002328
II, 20° , $0 \rightarrow 2^\circ$	ΔC_L	+0.101449	-0.007930	-0.000010	+0.093509
II, 20° , $0 \rightarrow 2^\circ$	D_L	-0.026344	-0.010408	-0.000003	-0.036755
II, 20° , $0 \rightarrow 2^\circ$	ΔC_M^A	+0.011164	+0.005658	-0.000002	+0.016820
II, 20° , $0 \rightarrow 2^\circ$	D_M^A	+0.007512	+0.007105	-0.000001	+0.014614

The fixed-surface share of the lift departure remains 62.4–63.0% across three tested baseline windows and the direct and symmetry-reflected zero-incidence solutions. The direct-state total varies more, from -0.1368 to -0.1050 , because the earliest baseline has appreciable curvature. About 63% therefore describes an allocation relative to these empirical trends: its stability supports the component ranking within the tested definitions, rather than establishing an experimental component fraction or causal share.

B. Pressure redistribution and spatial moment weighting

The different lift and moment rankings can be examined in the pressure distributions at the same two incidences used in the budget. At all three span stations, upper leading-edge suction strengthens while the suction minimum near the hinge weakens (Fig. 6). This combination is consistent with positive fixed-surface lift growth and loss of control lift over $0 \rightarrow 2^\circ$. Here, pressure recovery denotes the chordwise rise downstream of the hinge suction minimum; it does not imply an attached boundary layer. These two-state sections show ordinary pressure changes. Establishing which regions fall below their preceding loading trends additionally requires the baseline-state integrals.

The regional integrations provide that distinction (Fig. 7). The forward region still gains lift, but less than its preceding trend, while the intermediate region loses lift. Both regions forward of the hinge therefore contribute to the fixed-surface departure. The aft region also loses lift, giving negative departures in all three chordwise regions. A stronger leading-edge suction peak is thus compatible with reduced overall fixed-surface lift growth when the rest of its loading and its preceding trend are considered.

Geometric fore/aft increments differ slightly from the solver fixed/control increments because the hinge junction is assigned differently. In each partition, the sum of the integrated contributions includes the entire surface. Pressure

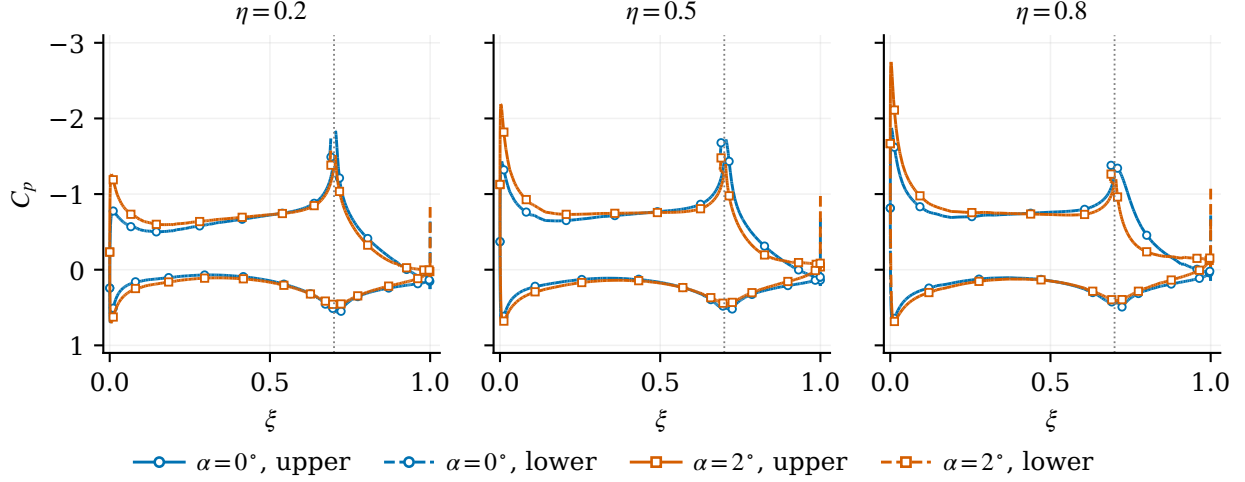


Fig. 6 Pressure sections for $I/\Gamma = 0^\circ/\delta = 20^\circ$ at direct $\alpha = 0, 2^\circ$ and undeformed $\eta = s/b = 0.2, 0.5, 0.8$. Circles/squares distinguish incidence; solid/dashed lines distinguish upper/lower sides. Sections follow the upper/lower fixed and control panels, with connectivity and face exclusions defined in Sec. II.D. The vertical dotted line marks the 70%-chord hinge. ξ is the local chord fraction obtained by undoing the hinge rotation.

supplies nearly all the principal lift departure; the small shear contribution is retained in the total, with effect-scale limits stated in Sec. II.D.

Comparing the aft regions along the span then explains why their moment contribution differs from their lift contribution. Similar aft lift losses produce moment increments about the fitted aerodynamic center that increase from about 0.0025 inboard to 0.0091 outboard. The force changes have different spatial weights in the moment integral, consistent with the importance of load location in finite-wing pitching response [3]. Here the evidence is the integrated regional change, rather than a continuous load density; Appendix A tests the reference and partition dependence.

C. Deflection contrast and reversal evolution

Increasing control deflection tests whether the reference distinction between ordinary growth and reduced growth persists in the earlier nonlinear interval. The $I/\delta = 30^\circ$ results in Table 5 again combine positive ordinary fixed lift growth with negative fixed and control departures. Its fitted-center moment contributions compete, however, and the total departure changes sign across tested windows. The lift-budget pattern extends to this deflection, while its moment response is more sensitive to the empirical preceding trend.

The associated reversal measure helps relate this budget to the changing aft pressure and loading. For $I/\delta = 20^\circ$, the upper-aft chordwise reversal fraction grows from 0.146 to 0.485 over direct $0 \rightarrow 2^\circ$, with reversal already present at -2° (Fig. 8). At $\delta = 30^\circ$, it grows from 0.622 to 0.935 over $-10 \rightarrow -8^\circ$. Expansion of an existing reversed-shear region accompanies reduced aft loading; the indicator describes this association without equating its area to every form of three-dimensional separation.

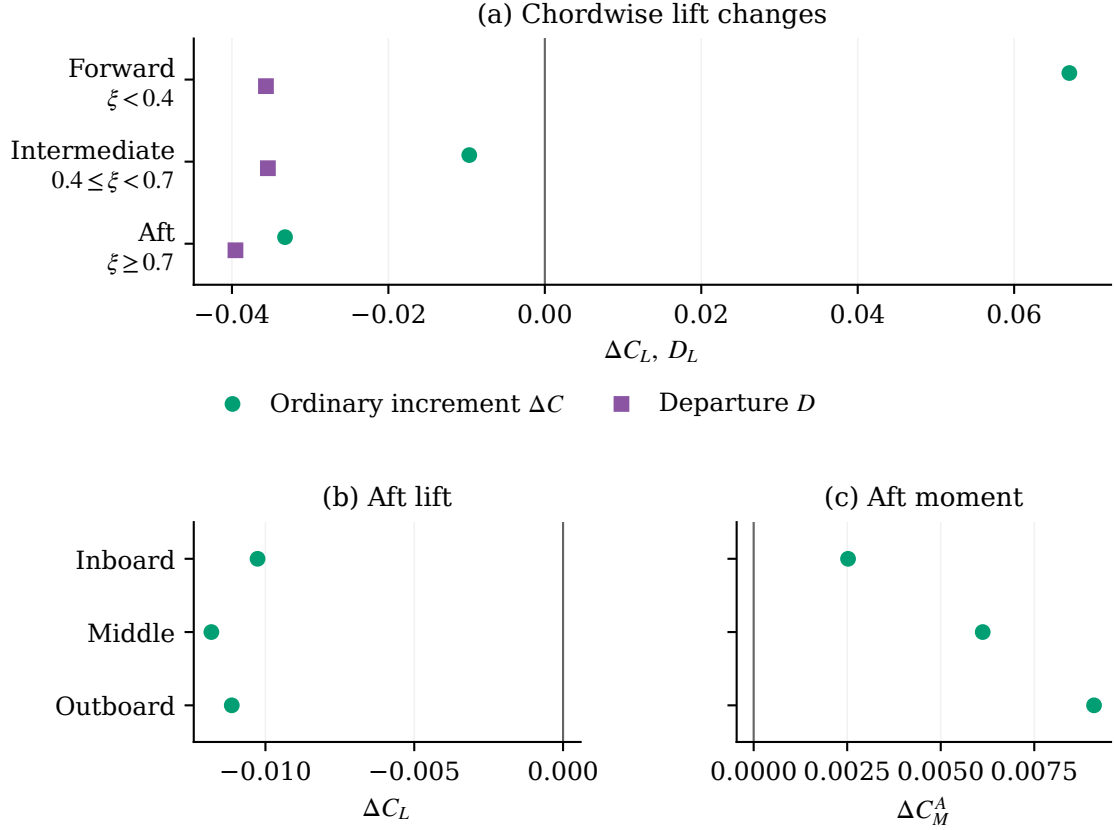


Fig. 7 Integrated geometric-region changes for $I/\Gamma = 0^\circ/\delta = 20^\circ$, direct $0 \rightarrow 2^\circ$: (a) paired chordwise lift increments and departures; (b,c) aligned ordinary aft lift and moment increments over inboard ($\eta < 1/3$), middle ($1/3 \leq \eta < 2/3$), and outboard ($\eta \geq 2/3$) regions. Baseline: $\{-6, -4, -2\}^\circ$; fitted axis: $x_A = 0.642$ m. Geometric regions follow Sec. II.D; separately retained endcaps complete the total beyond the displayed rows. Markers represent integrated contributions, rather than densities or solver surface groups.

The inclined solutions show why reversal area alone is insufficient to explain the integrated response. For $I/\Gamma = 45^\circ/\delta = 30^\circ$, the control is already extensively reversed over the selected moment interval. At later incidences the stored states are nonmonotonic: between 2 and 4°, the fraction falls from 0.579 to 0.090 at $\delta = 20^\circ$ and from 0.912 to 0.203 at $\delta = 30^\circ$. The available steady samples do not establish the origin of these changes. They also contain a distinct leading-edge reversed-shear region, for which the local transition-model fields give additional state information.

In the inclined $\delta = 30^\circ$, $\alpha = 2^\circ$ solution, this region is on the upper outboard leading edge ($2/3 \leq \eta \leq 0.95$, $0 \leq \xi \leq 0.15$). Modeled BCM intermittency on the reversing-centroid subset has median 2.4×10^{-5} and maximum 8.6×10^{-5} , with zero wall eddy viscosity and SA working variable. The low wall-model values characterize this particular stored feature alongside the aft reversal evolution. They do not provide a laminar/turbulent boundary-layer profile or establish a laminar separation bubble.

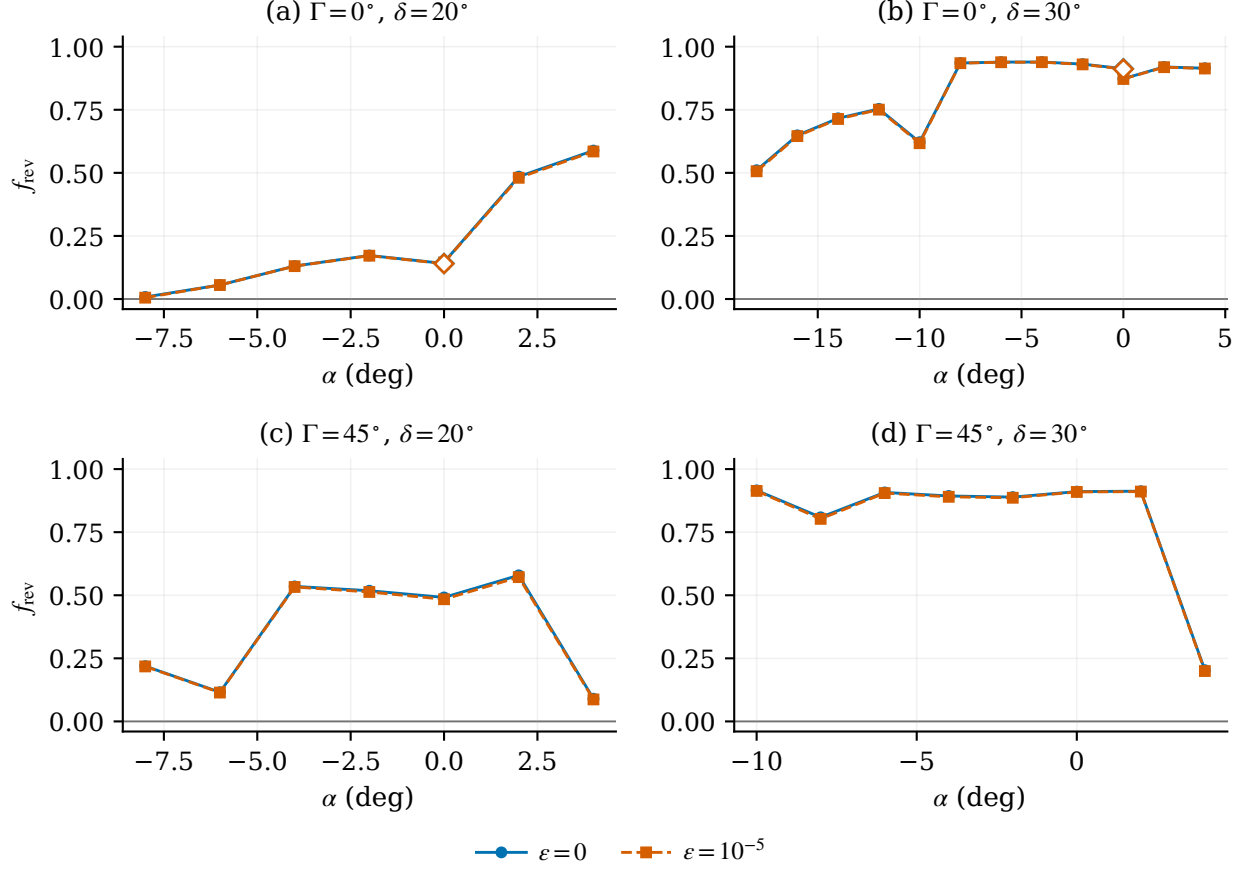


Fig. 8 Upper geometric-aft chordwise reversal fractions for I at both dihedral settings and $\delta = 20, 30^\circ$. Actual triangle area supplies the denominator. Circles/solid: $\epsilon = 0$; squares/dashed: $\epsilon = 10^{-5}$. Open diamonds identify symmetry-reflected zero-incidence solutions. All selected samples, including the nonmonotonic inclined 4° states, are shown. The indicator measures reversed chordwise wall shear over the specified region.

V. Dihedral and Planform Dependence

A. Matched dihedral: levels and incidence changes

The dihedral comparison examines how changing the spatial orientation of the same planform affects the joint loading response. Geom. I, δ , α , S , and MAC are held fixed; orientation and the associated root cut/domain change together. It therefore offers a controlled geometric contrast within the dataset, without supplying intermediate-angle or aircraft-level predictions.

The measurements first distinguish the size of the lift departure from its size relative to each model's preceding growth. At $\delta = 20^\circ$, absolute departures are smaller with dihedral, but over $-2 \rightarrow 0^\circ$ the descriptive relative deficits are 80.6% and 76.4% for $\Gamma = 0$ and 45° , whereas over $-2 \rightarrow 2^\circ$ they are 64.5% and 82.2%. These sampled ratios have no assigned significance level. A smaller absolute departure consequently does not imply uniformly weaker relative suppression.

The numerical force decomposition addresses the absolute level change before considering its evolution with incidence. At $\alpha = 0^\circ$, $\delta = 20^\circ$, projection of the zero-dihedral normal load contributes about -0.247 of the matched lift difference -0.301 . Changed local normal loading and span-force projection complete the balance. This coordinate accounting distinguishes projection from changed forces in the level difference, without isolating a causal contribution to nonlinear lift growth.

Moment requires a further comparison because dihedral changes both loading and its spatial weights. To examine how the moment evolves over the same incidence interval, rather than only comparing its absolute level, both numerical configurations are referred to the common axis A_I . Figure 9 evaluates

$$\mathcal{I}_M = [C_{M,45}^{A_I}(\alpha_b) - C_{M,45}^{A_I}(\alpha_a)] - [C_{M,0}^{A_I}(\alpha_b) - C_{M,0}^{A_I}(\alpha_a)] \quad (6)$$

with identical incidence intervals. Contrasts of the integrated $z dF_x$ and $-(x - x_{A_I})dF_z$ terms complete this ordinary-increment difference. For $\delta = 30^\circ$ over $-2 \rightarrow 0^\circ$, $\mathcal{I}_M = -0.0184$, with both terms negative. Their changing contributions across the other intervals show how force changes and spatial moment weighting enter the incidence-dependent response. This comparison uses no baseline subtraction and does not separate geometry from its effect on the computed flow.

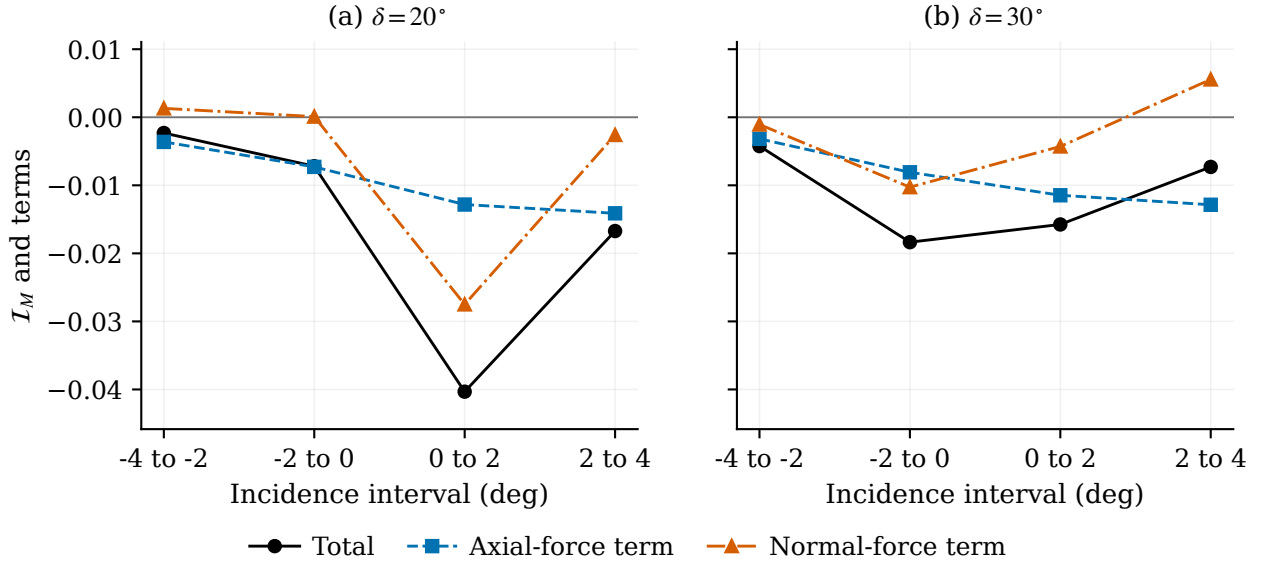


Fig. 9 Matched I dihedral contrasts of ordinary moment increments, 45° minus 0° , at $\delta = 20^\circ$ (a) and 30° (b). Moment uses common numerical axis $A_I = (0.642, 0, 0)$ m. The axial- and normal-force series are interval contrasts of the two normalized integrated terms in Eq. 5, whose sum is $\mathcal{I}_M$ in Eq. 6. Each marker represents its labeled 2° interval; lines guide the eye between interval summaries. Direct zero-incidence solutions are used, without baseline subtraction.

B. Second planform and scope of the result

The second planform tests whether the component relationship found for Geom. I also occurs with a different sweep, taper, aspect ratio, and chord distribution. For $II/\delta = 20^\circ$, Table 5 again shows positive ordinary fixed lift growth and negative control growth, with negative departures of both components. Its fitted-center moment departures are both positive and reinforce. A small partial cancellation about the original numerical origin has a different meaning: it is reference-specific and sensitive to baseline and zero-incidence choice (Appendix A). It therefore does not replace the fitted-center interpretation.

This recurrence is a result for the computed planform, whose measured lift slowdown is poorly reproduced (Sec. III). The direct and symmetry-reflected zero-incidence solutions remain distinct in Fig. 4, with their corresponding interval sensitivity in the appendix. Sweep, taper, aspect ratio, chord distribution, area, and MAC Reynolds number all differ between I and II. The comparison accordingly extends the component analysis to another planform without identifying an isolated sweep effect.

For the tested incidence response, the useful information is the combination of continued integrated lift growth, reduced growth of component loads, and their different moment weights. A lift curve or constant-slope description alone does not supply this allocation. Nonlinear models can admit loading changes through sectional data and corrections, as discussed in Sec. I; the present analysis supplies a spatially resolved account for the RANS states rather than a general verdict on reduced-order capability. The departures vary incidence at fixed deflection, not control deflection, and therefore do not measure a control derivative. Their engineering scope is the tested planforms, full-span control geometry, and conditions.

VI. Conclusions

The measured pre-stall behavior is a joint lift and pitching-moment response whose features need not appear at the same deflection or over the same incidence interval. Geom. I at zero dihedral exhibits a pronounced 20° lift slowdown, while a moment-response change is already present at 10° . The early 30° lift response is broader and its departure is window-sensitive, while the inclined moment features are not clearly resolved above both uncertainty sensitivities.

The reference numerical budget shows why continuing lift growth is compatible with a substantial loading departure. For $I/\Gamma = 0^\circ/\delta = 20^\circ$, fixed-surface lift still increases, but both fixed and control loads fall below their preceding trends. About 63% of the total lift departure is allocated to the fixed part across the tested windows and zero-incidence solutions, while its absolute amplitude is more baseline-sensitive. Increased leading-edge suction coexists with hinge/aft losses; regional integrations establish reduced expected growth in both fixed-chord regions. About the fitted aerodynamic center, the two moment departures reinforce and the control contributes more strongly. Similar aft lift losses also produce larger moment changes outboard, linking the different component rankings to spatial loading.

Changing orientation distinguishes lower absolute load from weaker relative suppression. The matched dihedral

analysis quantifies load projection at one incidence and the changing contributions of both moment-arm terms over common incidence intervals. The second planform also has reinforcing fitted-center moment departures, extending the component comparison without isolating its simultaneous geometric changes.

These interpretations describe the computed states. Their partial experimental correspondence includes substantial reference lift bias, inclined irregularities, and an inadequately reproduced second-planform lift slowdown. Future computations on the finer levels of the established mesh hierarchy are planned to assess the resolution sensitivity of the pre-stall load redistribution, with particular attention to pitching moment and the fixed- and control-surface contributions. In the reference computed case, fixed-surface lift growth can mask that surface’s substantial contribution to reduced total growth, while the control supplies the larger departure in pitching moment about the stated reference.

A. Sensitivity of the Numerical Loading Interpretation

A. Baseline and zero-incidence choices

Table 6 separates the absolute departure from its component allocation. The three unweighted windows use identical samples for every component, and direct and symmetry-reflected zero-incidence solutions are retained separately. Although the earliest pre-feature window has appreciable curvature, the positive fitted-center moment departure and fixed lift fraction persist as the total lift deficit varies.

Table 6 $I/\Gamma = 0^\circ/\delta = 20^\circ$, $0 \rightarrow 2^\circ$: **baseline and zero-incidence sensitivity.** D_M^A uses $x_A = 0.642$ m. **Fixed fraction is $D_{L,\text{fixed}}/D_{L,\text{total}}$, without an assigned confidence level.**

Zero solution	Baseline (deg)	Total D_L	Fixed fraction (%)	Total D_M^A
Direct	$\{-6, -4, -2\}$	-0.110605	62.55	+0.033238
Direct	$\{-8, -6, -4\}$	-0.136788	62.43	+0.040913
Direct	$\{-4, -2, 0\}$	-0.105014	62.64	+0.033958
Reflected	$\{-6, -4, -2\}$	-0.111790	62.76	+0.033493
Reflected	$\{-8, -6, -4\}$	-0.137973	62.60	+0.041168
Reflected	$\{-4, -2, 0\}$	-0.106792	62.98	+0.034341

The early $I/\delta = 30^\circ$ fitted-center total moment departure varies from approximately -0.0262 to $+0.0076$ across tested windows. For $I/\delta = 20^\circ$, the symmetry-reflected zero-incidence solution gives a $0 \rightarrow 2^\circ$ lift increment of 0.1121 instead of direct 0.0935 and a fitted-center moment departure of 0.0070 instead of 0.0146. Each coefficient and surface field retains its corresponding state; alternate zero solutions are not averaged and do not constitute a hysteresis measurement.

B. Spatial partition and reversal threshold

The whole fore/aft sums survive the tested span-third versus span-quarter partition, while finer chord bins preserve the two negative fixed-chord departures. Redefining the inboard aft boundary from $\eta < 1/3$ to $\eta < 1/4$ changes its lift departure from -0.01270 to -0.00939 . This measures partition sensitivity, rather than regional discretization error.

Exact matching to solver surface groups is available for six direct baseline states with complete facet coverage; the other spatial results use geometric bins.

For $I/\delta = 20^\circ$, the upper-aft reversal fraction at direct zero is 0.14578 versus 0.14089 at reflected zero. Threshold 10^{-5} changes the direct $0 \rightarrow 2^\circ$ sequence to 0.14497–0.48043, preserving its growth. Typical stored vector skin-friction norms are of order 10^{-3} , so the small thresholds assess near-zero sensitivity without a statistical uncertainty interpretation.

C. Moment-reference effects

At the original numerical origin, the baseline total moment departure is +0.10028, with fixed +0.05286 and control +0.04743. At the fitted aerodynamic center, the corresponding values are +0.03324, +0.01069, and +0.02255. The allocation changes while reinforcement persists. All reference transports use body $C_{F,z}$ in Eq. 2; the experimental comparison retains its own fitted center.

About the original numerical origin, the direct II fixed/control moment departures are $-0.00231/+0.00352$, with total +0.00121. This small cancellation is reference-specific and does not describe fitted-center reinforcement. For the baseline aft-span regions, the moment increments at the original origin are approximately 0.0082, 0.0129, 0.0156 from inboard to outboard. Their magnitudes differ from those at the fitted center, while the outboard weighting persists.

D. Saved-state and iterative documentation

Terminal histories match the force reports for 41 of the 44 interpreted surface states. The exceptions are $I/\Gamma = 0^\circ/\delta = 30^\circ$ at displayed $\alpha = -6^\circ$, and $II/\delta = 20^\circ$ at displayed $\alpha = -4^\circ$ and direct 0° . Complete final-stage reconstruction-order registration is unavailable for every state. For the inclined $\delta = 20^\circ$ irregularity over $-6 \rightarrow -4^\circ$, last-100 lift ranges are approximately 10^{-4} , much smaller than the between-state change. Terminal consistency and small iterative ranges do not establish steady-flow uniqueness or physical accuracy.

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